

■ Senior Citizens Chat Buddy

A modular, senior-friendly conversational AI system

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Code Repository:

<https://github.com/gitouser/SeniorChatBuddy/tree/main>

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A modular, senior-friendly conversational AI system

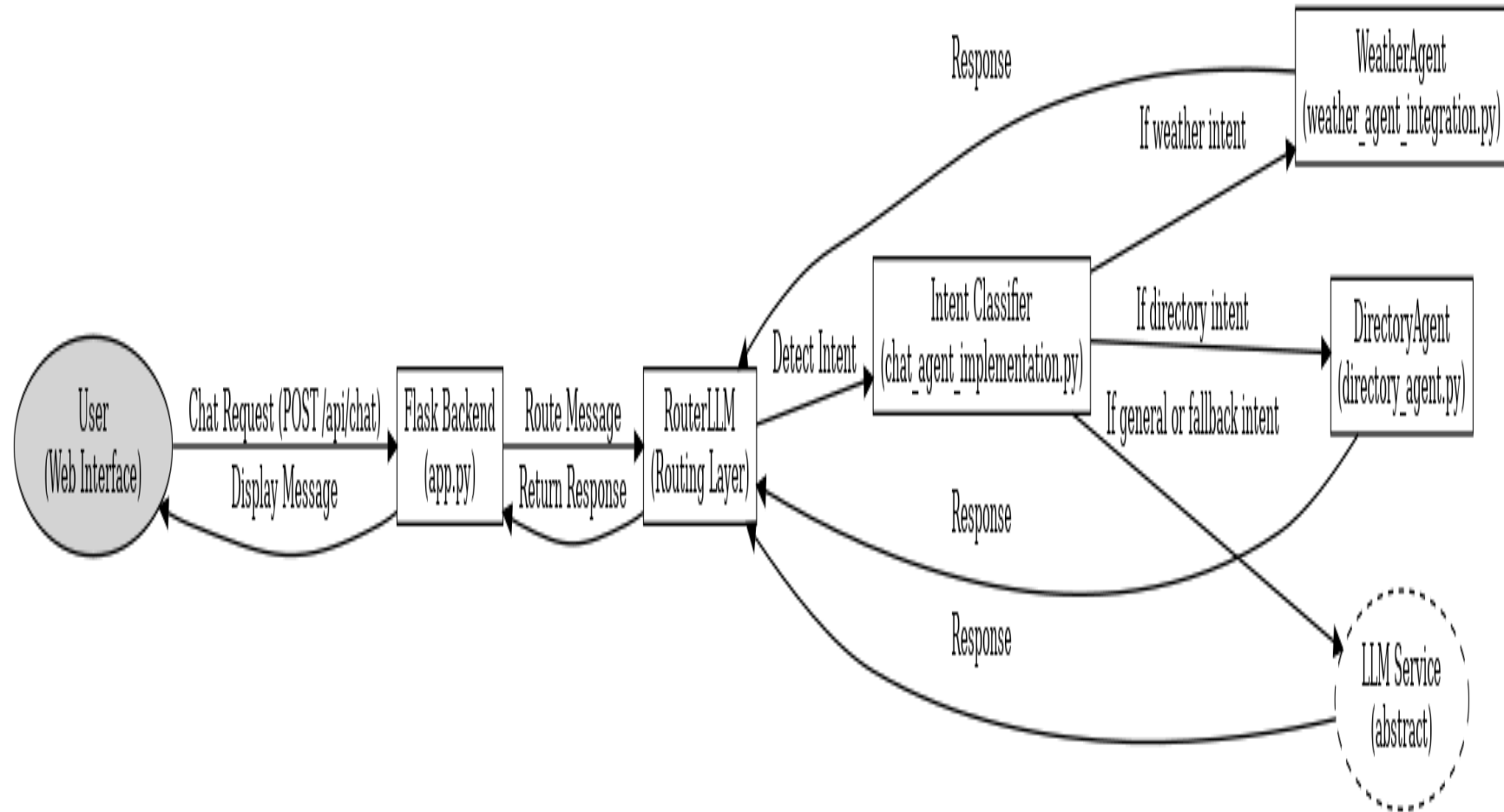
Goal:

To be a helpful assistant and conversation buddy for persons living alone or seniors, encouraging:

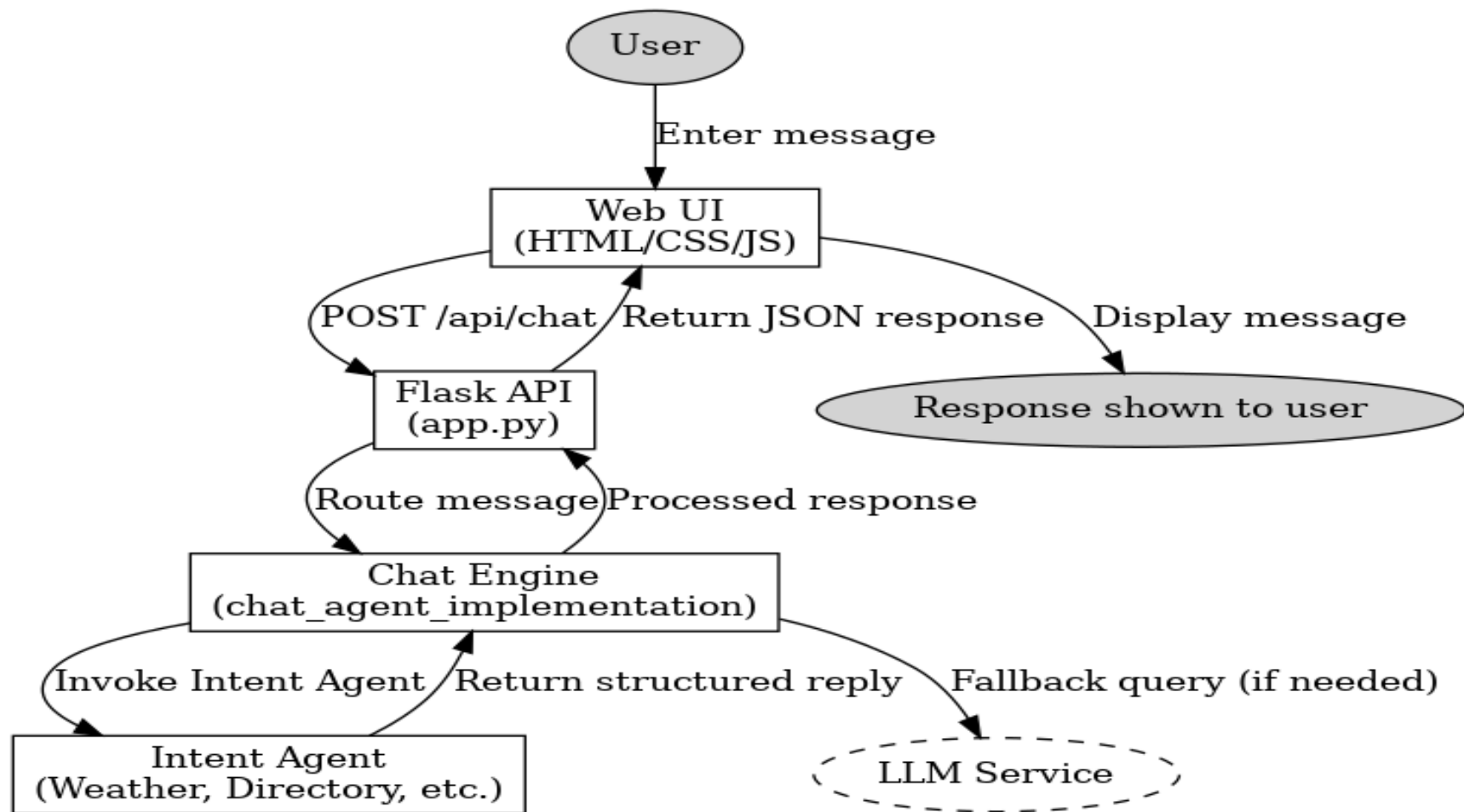
- Companionship
- Engagement
- Mental wellness

This project provides an accessible, web-based AI companion that uses a hybrid AI approach combining intent classification, topic-specific agents, and an LLM fallback. The design emphasizes simplicity, accessibility, and real-time, friendly interaction.

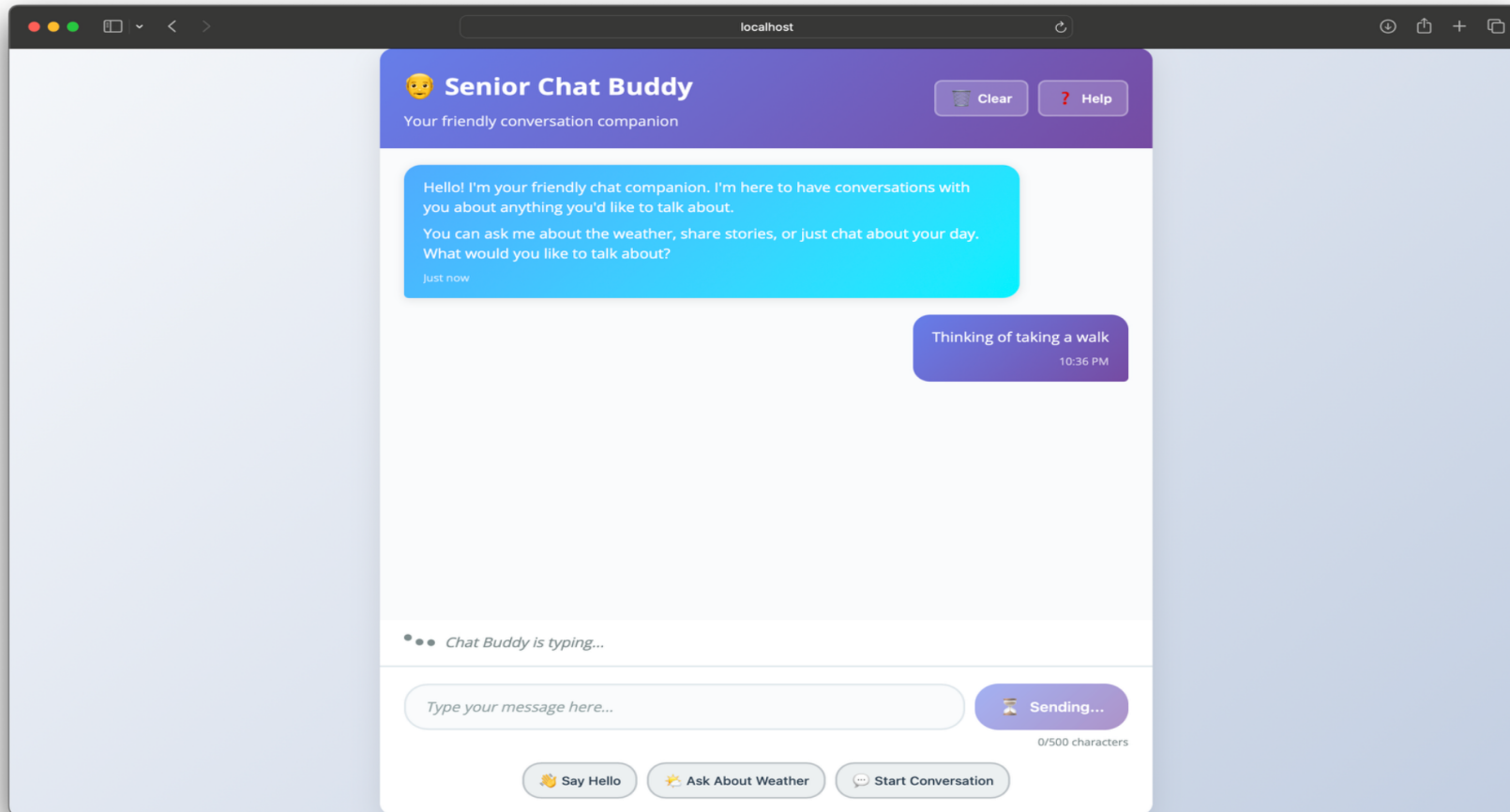
Technical Flowchart



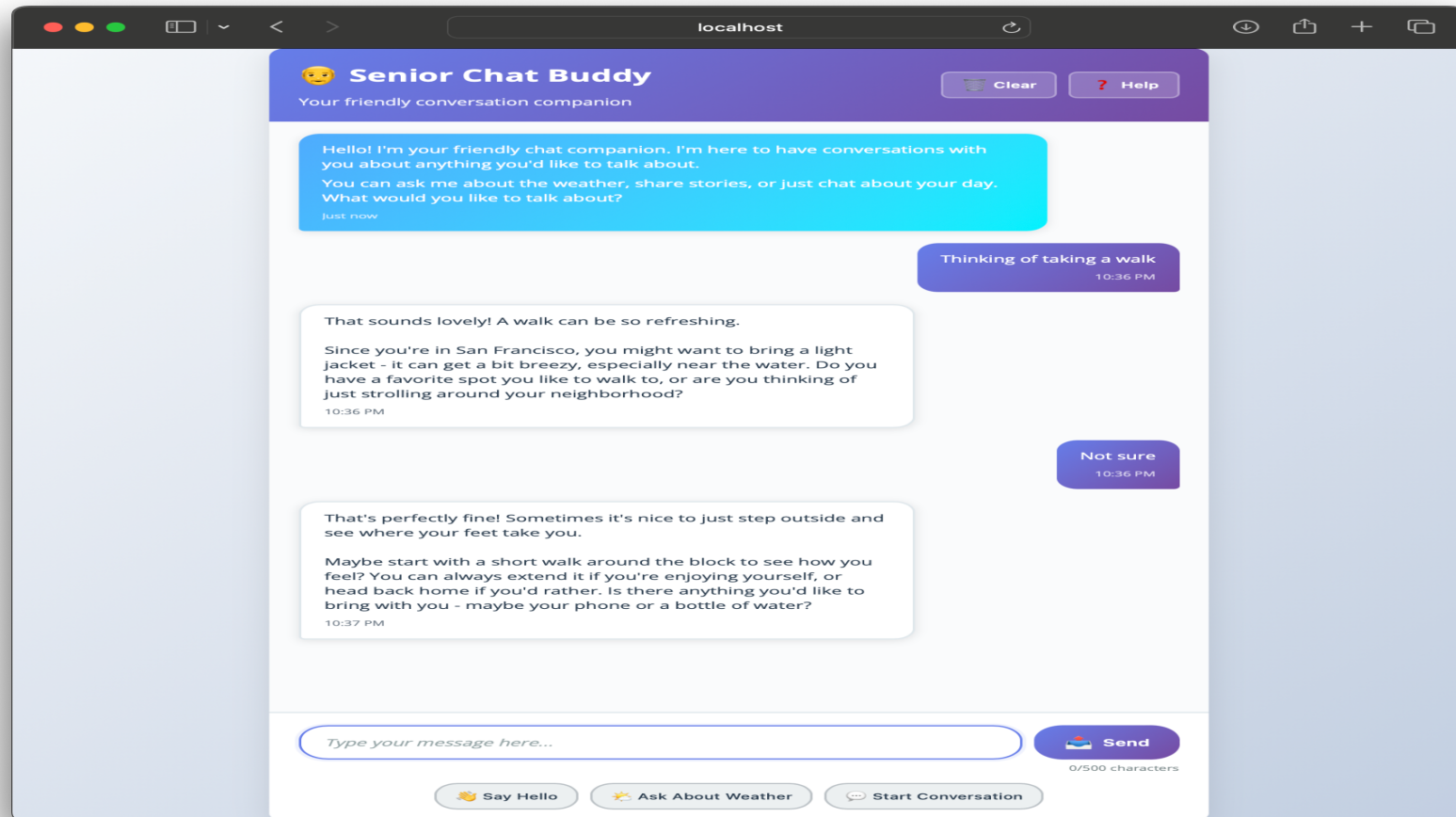
User Interaction Flow



Chat Interface – Welcome Screen



Chat Interface – Conversation in Progress



Interface Features & Accessibility

Top-right controls:

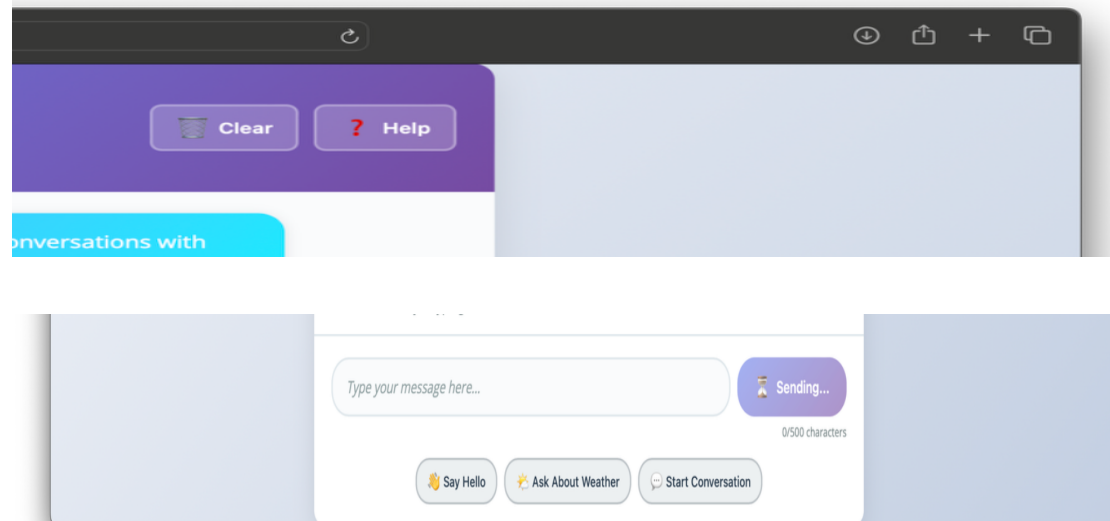
- **Clear** — clears both the chat history and the session context, allowing a completely new interaction.
- **Help** — provides easy-to-follow guidance on using the app, with example prompts and feature explanations.

Bottom action buttons:

- **Say Hello** — a friendly way to begin a conversation.
- **Ask about the Weather** — instantly invokes the Weather Agent for local updates.
- **Start Conversation** — opens general dialogue mode for casual, open-ended chats.

Accessibility Design:

- Senior-friendly large fonts (≥ 16 pt) and high-contrast colors for readability.
- Generous spacing and large touch targets (≥ 44 px) for ease of use.
- Clean, uncluttered layout minimizing cognitive load and enhancing comfort.



Solution Overview

- 1. Architecture:** Flask backend connects a web-based chat interface to multiple agents (Weather, Directory, etc.) via RouterLLM. RouterLLM leverages **LangChain** for advanced conversation management and storing conversation history.
- 2. Multi-turn Intent Handling:** The classic multi-turn intent issue is addressed through an intelligent LLM-based Router working in conjunction with a deterministic intent detection module.
- 3. Agents:** Topic-specific modules handle domain queries. WeatherAgent uses live data; DirectoryAgent searches local resources.
- 4. Accessibility:** Large fonts, clear buttons (“Clear”, “Help”, “Say Hello”), and simplified interaction ensure senior-friendliness.
- 5. Extensibility:** New intents and agents can be added easily by registering them in ChatAgent.
- 6. Testing & Deployment:** Full test suite covers intent detection, API routes, and integration. Deployable via Flask or Docker.

Social Impact

Empathy through AI

Social Impact:

- Addresses loneliness and accessibility barriers among seniors.
- Encourages companionship, conversation, and cognitive engagement.
- Designed for inclusivity with high-contrast visuals and simple interaction patterns.
- Pilot-ready for deployment in community or senior centers.

Next Steps & Enhancements

1. **Modular Expansion:** The architecture supports adding new agents easily — e.g., a **Calendar Agent** to read and manage user events.
2. **Voice Front-End:** The Flask interface can be replaced or complemented by a **voice-enabled front-end** for natural verbal conversations.
3. **Personalization:** Add light memory features to recall prior interactions and tailor responses.
4. **Wellness & Reminders:** Integrate smart reminders for medication, appointments, and daily activities.
5. **Multi-language & Accessibility:** Extend the interface to support multiple languages and voice synthesis for inclusivity.
6. **Usage & Feedback Dashboard:** Planned capability to measure engagement, conversation frequency, and sentiment trends for continuous improvement.

Repository Structure & Contributions

GitHub Repository:

<https://github.com/gitosuser/SeniorChatBuddy/tree/main>

Structure Overview:

SeniorChatBuddy/

■ ■ ■ ■ ChatBuddy/

■ ■ ■ ■ app.py

■ ■ ■ ■ chat_agent_implementation.py

■ ■ ■ ■ phi_router_llm.py

■ ■ ■ ■ weather_agent_integration.py

■

■ ■ ■ ■ Senior-Companion-Agent/

■ ■ ■ ■ directory_agent.py

■ ■ ■ ■ directory_agent_integration.py

■ ■ ■ ■ restaurant_and_pharmacy_check.py

■ ■ ■ ■ weather_assistant.py

■

■ ■ ■ ■ README.md

Contributors:

- **Aishwarya Ravisankar** — Implemented **Intent Agents** (Weather, Directory, etc.) and supporting logic.
- **Aparna Seetharaman** — Developed the **RouterLLM**, **ChatAgent**, and **Flask integration**, focusing on orchestration and system architecture.